Lightest full-fidelity feasible design per gas, u_res_max = 1.0 m/s — full run

Horizontal Displacement
x [m]

Horizontal Velocity
u_x [m/s]

Horizontal Acceleration
a_x [m/s^2]
-A_LIM

Horizontal Force
F_x [N]

Vertical Penetration
y_pen [m]
Displacement Along n

Vertical Velocity
u_y [m/s]
Velocity Along n

Vertical Acceleration
a_y [m/s^2]
-A_LIM
Bag Load Along n (F_n/M)

Net Vertical Force (incl. g)
F_y,net [N]
Pressure Force Along n

h [m]
dotted: D0 of each design

u_n [m/s]

F_n/M [m/s^2]
A_LIM

F_n [N]

Gas Mass
m [kg]
Time [s]

Airbag Volume
V [m^3]
Time [s]

Internal Pressure
1e5
P_bag [Pa]
Time [s]

Gas Temperature
T [K]
Time [s]

N2: cylinder / Nylon, m = 0.404 kg
He: cylinder / Nylon, m = 0.282 kg
Ar: cylinder / Nylon, m = 0.530 kg
CO2: cylinder / Nylon, m = 0.479 kg
dry air: cylinder / Nylon, m = 0.410 kg